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8. LINUX FILE SYSTEM ANALYSIS

AIM

To study the Linux file system and understand the organization of files, directories, permissions, and their role in system administration and security.

PROCEDURE

1. **Examine the overall structure and hierarchy of the Linux file system.**
2. **Learn the purpose of essential directories such as /etc, /var, /home, and /root.**
3. **Analyze file permissions, user ownership, and access control mechanisms.**
4. **Practice Linux commands for navigating, viewing, and managing files and directories.**
5. **Investigate suspicious files and detect possible security issues within the system.**

QUESTIONS

TASK 1 - Investigation Setup

After updating the `PATH` and `LD_LIBRARY_PATH` environment variables, run the command `check-env`. What is the flag that is returned in the output?

THM{5514ec4f1ce82f63867806d3cd95dbd8}

✓ Correct Answer

TASK 2 - Files, Permissions, and Timestamps

To practice your skills with the `find` command, locate all the files that the user **bob** created in the past 1 minute. Once found, review its contents. What is the flag you receive?

THM{0b1313afd2136ca0faafb2daa2b430f3}

✓ Correct Answer

Extract the metadata from the `reverse.elf` file. What is the file's MIME type?

application/octet-stream

✓ Correct Answer

Run the `stat` command against the `/etc/hosts` file on the compromised web server. What is the full **Modify Timestamp (mtime)** value?

2020-10-26 21:10:44.000000000 +0000

✓ Correct Answer

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TASK 3 - Users and Groups

Investigate the user accounts on the system. What is the name of the backdoor account that the attacker created?

b4ckd00r3d — ✓ Correct Answer

What is the name of the group with the group ID of **46**?

plugdev — ✓ Correct Answer

View the `/etc/sudoers` file on the compromised system. What is the full path of the binary that Jane can run as sudo?

/usr/bin/pstree — ✓ Correct Answer

TASK 4 - User Directories and Files

View Jane's `.bash_history` file. What flag do you see in the output?

THM{f38279ab9c6af1215815e5f7bbad891b} — ✓ Correct Answer

What is the hidden flag in Bob's home directory?

THM{6ed90e00e4fb7945bead8cd59e9fcd7f} — ✓ Correct Answer

Run the `stat` command on Jane's `authorized_keys` file. What is the full timestamp of the most recent modification?

2024-02-13 00:34:16.005897449 +0000 — ✓ Correct Answer

TASK 5 - Binaries and Executables

Run the `debsums` utility on the compromised host to check only configuration files. Which file came back as altered?

/etc/sudoers — ✓ Correct Answer

What is the `md5sum` of the binary that the attacker created to escalate privileges to root?

7063c3930affe123baecd3b340f1ad2c — ✓ Correct Answer

TASK 6 - Rootkits

Run *chkrootkit* on the affected system. What is the full path of the `.sh` file that was detected?

/var/tmp/findme.sh — ✓ Correct Answer

Run *rkhunter* on the affected system. What is the result of the `(UID 0) accounts` check?

Warning — ✓ Correct Answer

RESULT

Successfully analyzed the Linux file system and understood how file structure, permissions, and commands are used to manage and secure a Linux system.